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MIRAGE: Detecting Fake Emergency Electronic Brake Light Attacks in V2X Networks via Event-Gated Behavioral Analysis

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Abstract

The emergency electronic brake light (EEBL) application in Vehicle-to-Everything (V2X) networks warns upstream vehicles of hard braking. However, recent studies show that a compromised vehicle with valid credentials can broadcast fabricated EEBL messages, triggering unnecessary emergency braking and creating significant traffic safety risks. To detect such attacks, existing misbehavior detection systems apply general-purpose plausibility checks to every basic safety message, resulting in false alarms during normal driving and missed detections when the attacker generates a kinematically self-consistent trajectory. To address this problem, we propose MIRAGE (Multi-stage Inspection and Response Against Ghost-vehicle EEBL), an event-gated detector grounded in the Intelligent Driver Model (IDM). MIRAGE activates only under EEBL conditions, validates post-event trajectory consistency using position-derived speed, detects post-attack position freezing artifacts, and bounds the ego vehicle's response via IDM even on missed detections. We implement MIRAGE within the V2X Application Spoofing Platform (VASP) and evaluate against two fake EEBL attack variants alongside three existing defenses. MIRAGE achieves an F1 score of 0.922 across five random seeds with 88.3% detection rate at 0.9% false positive rate, and is the only detector that maintains high detection across both attack variants. We also show that MIRAGE reduces mean braking severity by 75%.

1 Introduction

Vehicle-to-Everything (V2X) communication enables vehicles to exchange real-time kinematic data via Basic Safety Messages (BSMs) at 10 Hz [22], supporting cooperative safety applications that extend awareness beyond the on-board sensor range [16]. Among these, the Emergency Electronic Brake Light (EEBL) is one of the first applications to be standardized for deployment: it notifies upstream vehicles of hard braking ahead, enabling earlier reaction when a direct line of sight is blocked [11, 35]. Given that rear-end crashes account

for a large fraction of multi-vehicle incidents [16], EEBL is recognized as one of the highest-impact V2X safety applications and is widely included as a Day-1 service in C-ITS deployment studies [35].

Every BSM is digitally signed under the vehicular Public Key Infrastructure (PKI), commonly the Security Credential Management System (SCMS), [49], which authenticates the sender as a legitimately enrolled vehicle [14, 19, 33], preventing external devices from injecting messages into the V2X channel. However, the PKI verifies only sender identity, not the contextual correctness of message content, so it cannot prevent an enrolled device from reporting fabricated position, speed, or brake status [7]. Independent verification is infeasible precisely because EEBL targets non-line-of-sight scenarios where receivers cannot directly observe the sender [37]. Prior work confirms the feasibility of these vectors through full remote attack chains on production vehicles [30], link-layer Controller Area Network (CAN) bus control from compromised ECUs [42], and hardware key extraction via voltage glitching [48]. Similarly, an industry report recorded 409 automotive cyber incidents in 2024, with 92% executed remotely [44]. An attacker who compromises a legitimate on-board unit (OBU) can pass PKI authentication using valid credentials and issue arbitrary BSM content under an apparently valid identity [49]. The attacker can forge brake flags, deceleration, and position fields to fabricate an EEBL event [5, 25]. The V2X Application Spoofing Platform (VASP) [5] rates the two EEBL-targeting attack variants implemented in its framework as high-risk attacks overall. In a related collaborative vehicular perception setting, Zhang et al. [52] demonstrated that crafted data can induce hard braking or increase collision risk, achieving over 86% success in high-fidelity simulation, with feasibility shown in real-world experiments. Together, these studies show that authenticated data-fabrication attacks are practical enough to require application-level validation beyond cryptographic authentication.

Existing Misbehavior Detection Systems (MDS) address content-level attacks through plausibility checks on BSM fields [7, 45], including range-and-threshold checks [6, 24],

cross-field physics validation [3, 36], and machine-learning (ML) classifiers trained on misbehavior datasets [15, 20, 21, 38, 46]. However, when applied to fake EEBL, these methods face two problems. First, always-on detectors, such as threshold checks [24] and F2MD [20], evaluate every BSM regardless of the application context. Sensor noise during normal car-following triggers false alarms [39], and at the application level, a false positive that suppresses a genuine EEBL warning may be more dangerous than the attack itself. Second, cross-field validators such as VCADS [3] detect inconsistencies across reported fields. However, VASP’s FakeEEBL_WithStop attack [5] produces a kinematically self-consistent stopping trajectory. This evasion strategy is well-established: the SixPack attack [32] and its successor [53] generate consistent fake braking trajectories across all six BSM kinematic fields to bypass cross-field validators.

We address these limitations with MIRAGE (Multi-stage Inspection and Response Against Ghost-vehicle EEBL), grounded in the Intelligent Driver Model (IDM). The detector activates only when EEBL trigger conditions are met, suppressing false alarms during routine driving. Upon activation, four stages evaluate contextual plausibility (Stage 1), deceleration consistency against physical and kinematic constraints (Stage 2), trajectory evidence through position-derived speed (Stage 3), and post-stop position consistency to catch frozen-position artifacts from artifact-bearing attack variants (Stage 4). The key insight is that position-derived speed, computed from consecutive position reports, serves as an independent kinematic witness that an attacker can suppress only by falsifying the position field. Failed checks accumulate into a weighted anomaly score for threshold-based detection, while an IDM-based mitigation layer bounds the ego vehicle’s braking response to limit the impact of missed detections. We evaluated MIRAGE within the VASP framework [5] against both attack variants and three baselines [3, 5, 20, 24]. MIRAGE achieves an F1 of 0.922 (true positive rate (TPR) = 0.883, false positive rate (FPR) = 0.009) and is the only detector maintaining high rates across both types.

In summary, we make the following contributions:

- We present an *integration of event-gated, multi-stage behavioral inspection with IDM-based response bounding* for fake EEBL attack defense. The contribution is this integration rather than any single check in isolation. The detector reuses established ingredients (plausibility checks, position-derived speed, post-event trajectory analysis), but couples them to an application-specific event gate that activates only under EEBL conditions and to a controller-level response layer that bounds the ego vehicle’s deceleration even when detection fails.
- We design the *event-gated, multi-stage detector* with four verification stages: contextual relevance of the EEBL claim (Stage 1), kinematic plausibility of the reported deceleration (Stage 2), post-event trajectory consistency through

position-derived speed (Stage 3), and post-stop position consistency (Stage 4). The architecture treats Stages 1–3 as defense-in-depth against kinematically inconsistent attacks, while Stage 4 is a targeted check against attack variants that produce kinematically self-consistent BSMs during the braking phase.

- We implement and evaluate MIRAGE within the *VASP framework* [5], benchmarking against three baselines under two fake EEBL attack variants. MIRAGE achieves an F1 of 0.922 with 77.0% $\mathcal{A}_1$ and 99.7% $\mathcal{A}_2$ detection rates; the $\mathcal{A}_2$ result reflects VASP’s current FakeEEBL_WithStop implementation and is discussed as a primary limitation in Section 8.3. The IDM bounding layer reduces peak braking severity by 50% compared to unprotected emergency braking, limiting deceleration to -4.0 m/s^2 even for undetected attack BSMs.

MIRAGE is publicly available, including the code, SUMO scenario files, Veins configurations, analysis scripts, and random seeds for reproducibility, and is archived at <https://doi.org/10.5281/zenodo.19639252> (see Section 10).

2 Background

2.1 EEBL Application

Emergency Electronic Brake Light (EEBL) is standardized in SAE J2945/1 [35] and is widely listed as a Day-1 C-ITS safety service. When a vehicle decelerates beyond a hard-brake threshold (SAE J2945 specifies $\geq 0.4g \approx 3.92 \text{ m/s}^2$; VASP’s implementation uses 4.5 m/s^2), it sets the `eventHardBraking` flag in its BSM [35]. A receiver triggers an EEBL warning if three conditions hold: (1) the sender’s brake flag is set, (2) the sender is longitudinally ahead and in the same or adjacent lane, and (3) the sender is within a safety-relevant distance [5, 18]. This protocol extends hazard awareness beyond direct line of sight, enabling earlier braking reactions and helping prevent chain-reaction rear-end collisions [40]. As receivers verify only the sender’s PKI enrollment rather than the physical existence of the claimed vehicle [25, 49], EEBL is vulnerable to ghost-node attacks in which an adversary creates a virtual entity that broadcasts fabricated hard-brake BSMs from a position where no real vehicle exists. The VASP platform [5] implements two types of EEBL attacks, formalized as $\mathcal{A}_1$ and $\mathcal{A}_2$:

- $\mathcal{A}_1$ (**FakeEEBL_NoStop**): A ghost vehicle j^* appears ahead of victim i (offset $d_0 = 60 \text{ m}$), sets $\beta_{j^*} = 1$, and reports $a_{j^*} = -5 \text{ m/s}^2$. The reported speed decreases to half the claimed rate while the position remains at constant velocity, so the position field implies that the ghost continues moving rather than decelerating as claimed.
- $\mathcal{A}_2$ (**FakeEEBL_WithStop**): A ghost vehicle appears ahead ($d_0 = 55 \text{ m}$) and decelerates at $a_{j^*} = -3.5 \text{ m/s}^2$ to

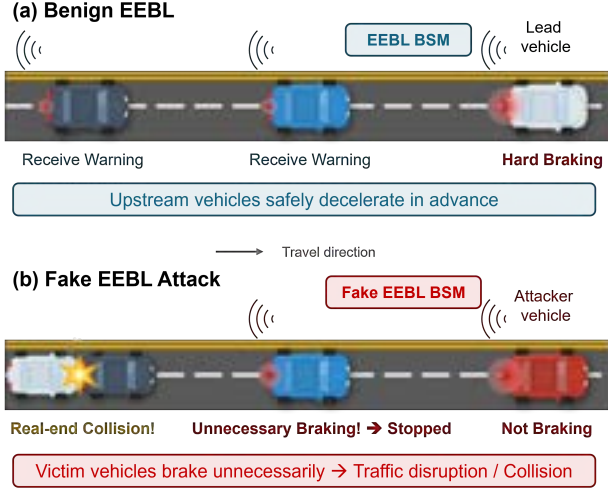


Figure 1: Comparison between legitimate EEBL protocol operation (top) and a fake EEBL attack (bottom).

a full stop, with all BSM fields kept mutually consistent so the trajectory is kinematically self-consistent and is the hardest case for BSM-only detection [32].

Figure 1 illustrates the contrast between benign EEBL operation and a fake EEBL attack. In the benign case (top), the leading vehicle undergoes genuine hard braking and broadcasts an EEBL BSM; upstream vehicles receive the warning and decelerate safely in advance, preventing a chain-reaction collision. In the attack case (bottom), an attacker broadcasts a fabricated hard-brake BSM while continuing at constant speed. The victim vehicle brakes unnecessarily and comes to a stop, while a following vehicle that cannot react in time collides with it, demonstrating that a single fake EEBL can cascade into real physical harm.

The ego vehicle runs an on-board MDS that processes incoming BSMs in real time. The defender has access only to BSM content (no radar/lidar confirmation in the non-line-of-sight (NLOS) scenario), must prioritize specificity since suppressing a genuine EEBL could cause a collision [45], and must process each BSM within one BSM interval ($\Delta t = 0.1$ s); the time from the first attack BSM to first detection is a separate quantity reported in Section 8.

2.2 Intelligent Driver Model (IDM)

The Intelligent Driver Model (IDM) [43] is a continuous-time car-following model that computes the acceleration of a following vehicle i as a function of its current speed, the gap to the leader, and the closing speed:

$$a_{\text{IDM}} = a_{\text{max}} \left[1 - \left(\frac{v_i}{v_0} \right)^\delta - \left(\frac{s^*(v_i, \Delta v_{ij})}{s_{ij}} \right)^2 \right], \quad (1)$$

where v_0 is the desired speed, a_{max} the maximum acceleration, δ the acceleration exponent (typically 4), s_{ij} the current gap to leader j , $\Delta v_{ij} = v_i - v_j$ the closing speed, and $s^*(v, \Delta v) = s_0 + vT + \frac{v\Delta v}{2\sqrt{a_{\text{max}}b}}$ is the desired minimum gap with jam distance s_0 , safe time headway T , and comfortable deceleration b .

The IDM has two properties that make it suitable for MIRAGE. First, its output is *gap-dependent*: the deceleration scales with the current gap s_{ij} and closing speed Δv_{ij} , so the ego's response naturally adapts to the actual traffic situation rather than applying a fixed emergency deceleration. When a claimed braking event is far away or the closing speed is low, the IDM prescribes mild deceleration; only when the gap is genuinely small does the output approach the comfortable deceleration limit b . Second, the model has been extensively calibrated against empirical traffic data and is widely adopted in traffic simulation [23, 43], ensuring that the mitigation layer's behavior is consistent with real driving dynamics. We use the IDM in two roles: (1) to generate realistic platoon dynamics in the Simulation of Urban MObility (SUMO) simulation [23, 26], and (2) to bound the ego vehicle's response to a claimed EEBL event (Section 5.7), with calibration parameters detailed in Section 6.2.

2.3 Time-to-Collision (TTC)

Time-to-Collision (TTC) [17] is a surrogate safety metric widely used in collision warning design [18] that estimates the time remaining before vehicle i would reach the position of leader j at the current closing speed:

$$\text{TTC}_{ij}^{(t)} = \begin{cases} \frac{s_{ij}^{(t)}}{\Delta v_{ij}^{(t)}}, & \text{if } \Delta v_{ij}^{(t)} > 0 \\ \infty, & \text{otherwise.} \end{cases} \quad (2)$$

TTC plays two roles in MIRAGE. First, it serves as a *context filter*: a genuine hard-brake event produces low TTC for nearby followers, whereas a ghost vehicle placed far ahead or receding from the ego yields high or undefined TTC, indicating that the claimed emergency lacks physical urgency. Second, the sign of the closing speed conveys directionality: $\Delta v_{ij} \leq 0$ indicates that the ego is not closing on the sender, so the EEBL has no safety relevance regardless of whether it is genuine. We exploit both properties in Check 1a (Section 5.2).

3 Threat Model

We follow the same threat model as in the V2X security literature [7, 28, 37], adapted to fake EEBL attacks.

The adversary aims to trigger false EEBL warnings that cause unwarranted hard-braking responses from victim vehicles. A successful attack degrades safety by inducing sudden deceleration and increasing rear-end collision risk, disrupts

traffic by propagating braking shockwaves through the platoon [2], and reduces trust in V2X warnings if false events occur repeatedly [31].

We consider an internal, authenticated adversary consistent with the taxonomy in prior work [7, 28]. The attacker holds legitimate V2X certificates obtained through a compromised OBU [9, 27], supply-chain compromise during device provisioning [31], or insider access at fleet operators [37]. Firmware-level exploits have been demonstrated on production vehicles, from remote wireless-to-CAN-bus attack chains [30] to link-layer message manipulation by software-compromised ECUs [42]. With this level of access, the attacker can set attacker-chosen values in BSM fields, including position $\mathbf{x}_j$, speed v_j , acceleration a_j , and brake flag β_j [5]. The attacker can inject a ghost vehicle, a virtual entity broadcasting BSMs from a position where no physical vehicle exists [14, 25]. The attacker operates within a typical Dedicated Short-Range Communications (DSRC) communication range (≤ 300 m [22]) and broadcasts at the standard 10 Hz rate. Sybil attacks, RF jamming, and combined multi-vector attacks are outside the scope and have been addressed by a separate line of V2X security work: foundational eviction and revocation mechanisms [34] were subsequently formalized into the Security Credential Management System (SCMS) [49] and have been revisited in recent V2X cybersecurity surveys [37], which document the gap that content-level misbehavior detection (the focus of this paper) is designed to fill.

The insider threat described above is not hypothetical. The VASP threat analysis rates the two EEBL-targeting attack variants implemented in its framework as *high risk* overall [5]. Data fabrication attacks on collaborative vehicular perception have achieved over 86% success in high-fidelity simulation, can induce hard braking or increased collision risk, and have been shown feasible in real-world experiments [52]. Meanwhile, misbehavior detection standards do not yet cover all V2X services [8, 29]: vehicular PKI authenticates senders but does not validate message content [49], as receivers lack an independent channel to verify the sender’s physical state in the non-line-of-sight conditions where EEBL operates [37]. An insider with valid certificates can pass all cryptographic checks yet still fabricate EEBL events. A gap remains between deployed authentication and content-level validation.

The attacker is aware of the BSM format and EEBL trigger conditions per SAE J2735/J2945 [35], common MDS approaches such as range checks and speed/acceleration bounds [20, 24], and the victim’s approximate position and speed from overheard BSMs. This knowledge enables the attacker to craft attacks that remain within plausibility bounds and to place the ghost vehicle at a realistic distance ahead. In this paper, the attacker conducts two primary attacks, $\mathcal{A}_1$ (FakeEEBL_NoStop) and $\mathcal{A}_2$ (FakeEEBL_WithStop), introduced in the VASP framework [5].

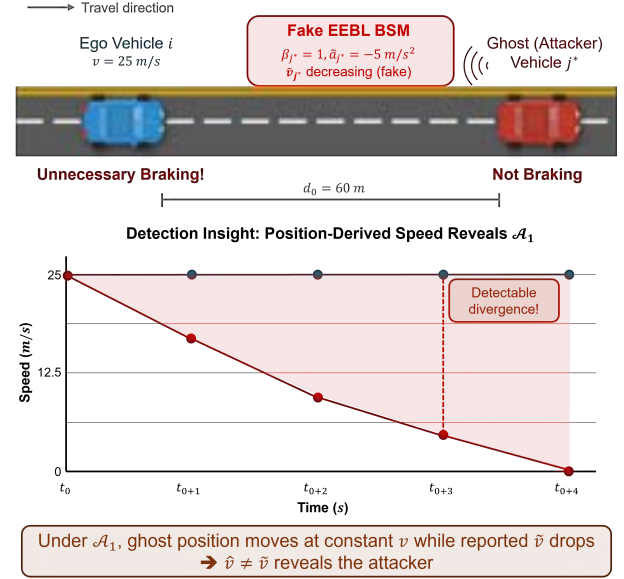


Figure 2: Motivating example under $\mathcal{A}_1$. Position-derived speed $\hat{v}_{j^*}$ remains near 25 m/s while reported $\tilde{v}_{j^*}$ drops to zero, revealing a detectable divergence.

4 Motivating Example

We illustrate the attack and the detection challenge through a concrete real-world scenario (shown in Figure 2). Ego vehicle i travels at $v_i^{(t_0)} = 25$ m/s. At time t_0 , a ghost vehicle j^* begins broadcasting BSMs claiming a position 60 m ahead, with $\beta_{j^*} = 1$ and $\tilde{a}_{j^*} = -5$ m/s². Vehicle i has no prior BSM history for j^* . All EEBL trigger conditions are met. A threshold detector (B1 in Section 6.3) verifies $|\tilde{a}_{j^*}| < 11$ m/s² and $\tilde{v}_{j^*} < 70$ m/s. Since the manipulated values remain within normal ranges, the detector does not flag the BSM. A cross-field detector works better under $\mathcal{A}_1$: it computes $\hat{a}_{j^*}^{(t)} = (\tilde{v}_{j^*}^{(t)} - \tilde{v}_{j^*}^{(t-1)})/\Delta t$ and finds $\hat{a}_{j^*} \approx -2.5$ vs. $\tilde{a}_{j^*} = -5$ m/s², yielding a discrepancy of 2.5 m/s² that meets the detection threshold τ_a . However, under $\mathcal{A}_2$, all reported fields satisfy kinematic equations, and the detector finds nothing anomalous.

The key observation is that *position-derived speed* $\hat{v}_{j^*}^{(t)} = |\mathbf{x}_{j^*}^{(t)} - \mathbf{x}_{j^*}^{(t-1)}|/\Delta t$, computed from consecutive position reports, reveals the attacker under $\mathcal{A}_1$. Since the ghost’s position continues at constant velocity even as $\tilde{v}_{j^*}$ drops, $\hat{v}_{j^*}$ remains near 25 m/s while $\tilde{v}_{j^*}$ approaches zero. The attacker cannot suppress this divergence without faking a consistent stopping trajectory, which requires $\mathcal{A}_2$ -level sophistication. We note that this check derives speed solely from the position field, bypassing the reported speed and acceleration fields. It is robust to manipulation of speed and acceleration in particular. However, since position is itself a reported BSM field, an attacker who controls position evolution consistently (as in $\mathcal{A}_2$) can defeat this check during the braking phase. We address this

through a post-stop position-consistency check (Stage 4) that detects frozen-position artifacts after the attack completes, complemented by an IDM mitigation layer that bounds the ego's response regardless of the detection outcome.

Finally, gating these checks to EEBL-active periods avoids false alarms during routine driving, where transient sensor noise would trigger cross-field checks on every BSM.

5 MIRAGE: Multi-stage Inspection and Response Against Ghost-vehicle EEBL

5.1 Overview

Figure 3 illustrates the end-to-end detection pipeline of MIRAGE. Unlike always-on MDS [20, 24, 45], MIRAGE performs deep inspection only when an incoming BSM from sender j satisfies two conditions. First, $\beta_j^{(t)} = 1$ and the sender is within communication range, $|\mathbf{x}_j^{(t)} - \mathbf{x}_i^{(t)}| \leq R_{\max} = 300$ m (regardless of relative direction; directional context is assessed in Stage 1). Second, at least two consecutive BSMs from the same sender carry the brake flag (the *consecutive-brake requirement*). The consecutive-brake filter ensures that a single transient brake flag, which may occur during normal car-following deceleration in SUMO, does not open the gate. We call this the **EEBL gate**. The gate remains open for up to $T_{\text{gate}} = 5$ s after the initial trigger, allowing post-event checks to accumulate evidence; the spatial and directional context of the EEBL event is then assessed in Stage 1. When the gate is open, four stages of evidence are evaluated. Each failed check k contributes a calibrated weight w_k to a cumulative anomaly score $S_j^{(t)}$. A BSM is flagged as malicious when $S_j^{(t)} \geq \Theta$. Table 1 summarizes the notation used throughout the paper.

5.2 Stage 1: Context Assessment

When a BSM passes the EEBL gate, the detector evaluates if the traffic context is consistent with a genuine emergency.

Check 1a (TTC plausibility). A genuine EEBL event typically involves low TTC for nearby followers. Van der Horst [47] identified a TTC of 4 s as the boundary between comfortable and uncomfortable car-following situations, a value widely adopted in collision warning design. We use $T_{\text{warn}} = 4$ s and flag when TTC exceeds twice this value:

$$\text{TTC}_{ij}^{(t)} > 8 \text{ s} \implies \text{flag} \quad (w_{1a} = 0.15). \quad (3)$$

If the ego is more than 8 seconds from reaching the claimed braking vehicle, the EEBL context lacks urgency. A stronger variant (Check 1a') flags when the sender is not longitudinally ahead of the ego *or* when $\text{TTC}_{ij}^{(t)} < 0$ (i.e., the sender is moving faster), indicating the EEBL has no safety relevance ($w_{1a'} = 0.35$). Checks 1a and 1a' are mutually exclusive.

Table 1: Summary of notation.

Symbol	Description
i	Ego (detecting) vehicle
j, j^*	Sender vehicle; attacker / ghost vehicle
$t, \Delta t$	Discrete time step, BSM interval (0.1 s)
$\mathbf{x}_j^{(t)}, v_j^{(t)}, a_j^{(t)}$	Position, speed, acceleration of j at t
$\hat{v}_j^{(t)}, \hat{a}_j^{(t)}$	Reported speed and acceleration in BSM
$\hat{v}_j^{(t)}$	Position-derived speed: $ \mathbf{x}_j^{(t)} - \mathbf{x}_j^{(t-1)} /\Delta t$
$\hat{a}_j^{(t)}$	Speed-derived accel.: $(\hat{v}_j^{(t)} - \hat{v}_j^{(t-1)})/\Delta t$
$\beta_j^{(t)}$	Hard-brake flag ($\beta_j = 1$: braking event)
$s_{ij}^{(t)}$	Gap between i and j : $ \mathbf{x}_j^{(t)} - \mathbf{x}_i^{(t)} $
$\Delta v_{ij}^{(t)}$	Closing speed: $v_i^{(t)} - v_j^{(t)}$
$\text{TTC}_{ij}^{(t)}$	Time-to-collision (Eq. 2)
$\mathcal{H}_j$	BSM history buffer for sender j
$S_j^{(t)}$	Anomaly score for j at time t
w_k, Θ	Weight of check k ; detection threshold
T_{gate}	EEBL gate timeout (5 s)
$\mathcal{A}_1, \mathcal{A}_2$	FakeEEBL_NoStop, FakeEEBL_WithStop

Check 1b (New-sender detection). A ghost vehicle that appears and claims hard braking has minimal BSM history. We flag if the number of previously received BSMs satisfies:

$$|\mathcal{H}_j| \leq N_{\min} = 3. \quad (4)$$

In normal traffic, a vehicle is observed for several seconds before a braking event [41]. A sender with ≤ 3 prior BSMs (≤ 0.3 s of history at 10 Hz) that immediately triggers EEBL is anomalous ($w_{1b} = 0.20$). We note that a legitimate vehicle entering communication range during a genuine hard brake would also trigger this check; however, since $w_{1b} = 0.20 < \Theta = 0.55$, this alone does not cause a false positive.

5.3 Stage 2: Deceleration Plausibility

Check 2a (Physical limit). On dry pavement, standard vehicles achieve peak sustained braking of approximately 0.8–0.9g [13]; the American Association of State Highway and Transportation Officials (AASHTO) road design uses a conservative deceleration rate of 3.4 m/s² ($\approx 0.35g$) for stopping sight distance calculations [1]. Deceleration at or above 1g is unattainable without specialized surfaces, making it physically implausible for standard vehicles:

$$|\tilde{a}_j^{(t)}| > g = 9.81 \text{ m/s}^2 \implies \text{flag} \quad (w_{2a} = 0.20). \quad (5)$$

Check 2b (Cross-field consistency). We compare speed-derived acceleration $\hat{a}_j^{(t)}$ against reported $\tilde{a}_j^{(t)}$:

$$|\hat{a}_j^{(t)} - \tilde{a}_j^{(t)}| > \tau_a = 2.5 \text{ m/s}^2 \implies \text{flag} \quad (w_{2b} = 0.15). \quad (6)$$

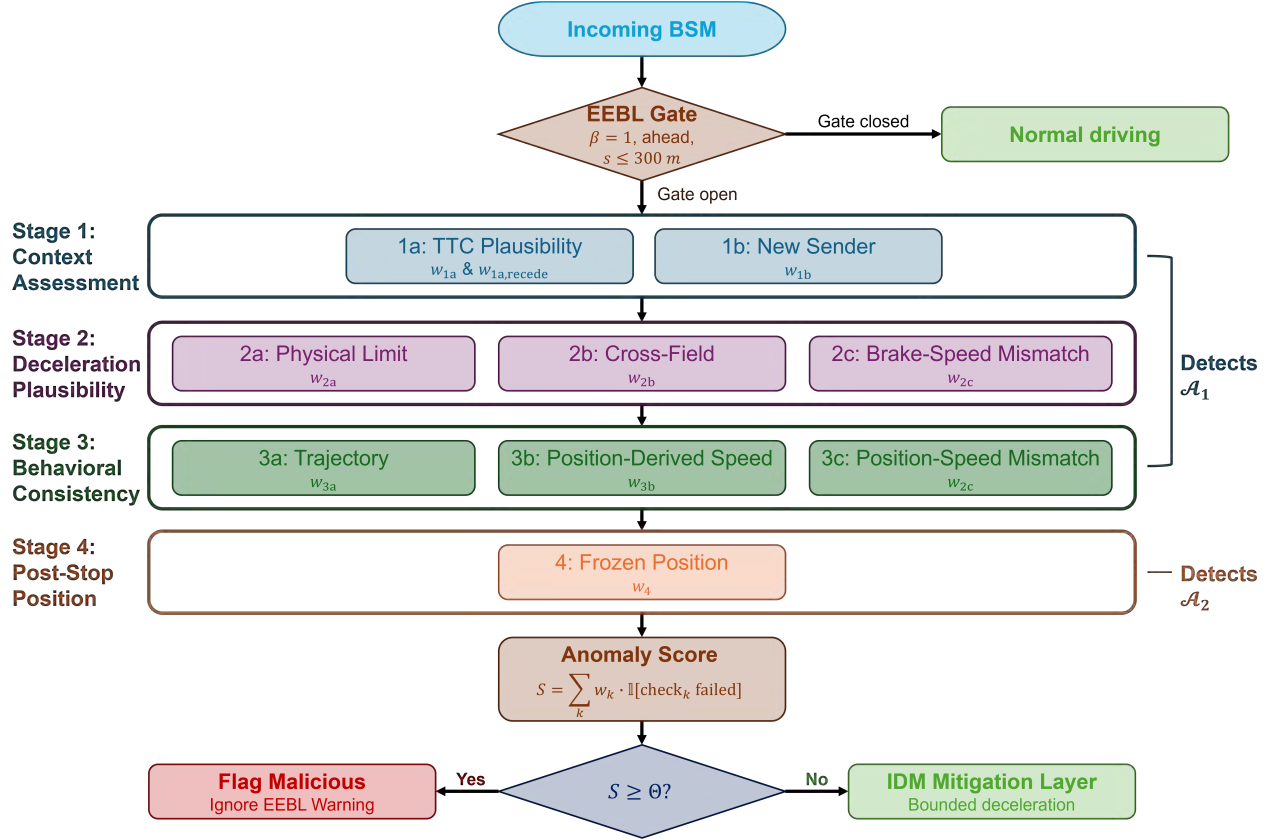


Figure 3: Overview of MIRAGE: Stages 1–3 target $\mathcal{A}_1$; Stage 4 targets $\mathcal{A}_2$; the IDM mitigation bounds the ego vehicle’s response on missed detections.

The tolerance τ_a accommodates SUMO’s discrete-time integration artifacts, which produce per-step acceleration discrepancies exceeding the nominal sensor noise ($\sigma_a = 0.15 \text{ m/s}^2$). This check catches $\mathcal{A}_1$, where reported speed drops at half the claimed rate, but not $\mathcal{A}_2$, where $\hat{a}_j \approx \tilde{a}_j$ by construction.

Check 2c (Brake-speed mismatch). If $\beta_j = 1$ but speed-derived deceleration is mild:

$$\beta_j^{(t)} = 1 \wedge |\hat{a}_j^{(t)}| < \frac{1}{2} a_{\text{hard}} = 1.96 \text{ m/s}^2 \implies \text{flag}, \quad (7)$$

where $a_{\text{hard}} = 0.4g = 3.92 \text{ m/s}^2$ is the SAE J2945 hard-brake threshold [35] ($w_{2c} = 0.15$). Neither $\mathcal{A}_1$ nor $\mathcal{A}_2$ consistently triggers this check, as both attacks reduce reported speed sufficiently to exceed the half-threshold. However, it provides defense-in-depth against simpler attack variants, such as a naive attacker that sets the brake flag to trigger EEBL without modifying the speed or acceleration fields. Because $w_{2c} = 0.15 < \Theta$, it contributes only as supporting evidence alongside other checks.

5.4 Stage 3: Behavioral Consistency

This stage evaluates post-event trajectory evidence, which is a capability that general-purpose per-BSM detectors do not

leverage. Stage 3 requires at least $N_{\min} = 3$ received BSMs per sender to activate, because position-derived speed requires two consecutive positions and trajectory consistency requires a baseline kinematic state; Check 3a then operates on a sliding window of $W = 6$ BSMs, and Check 4 (Stage 4) uses $N_{\text{freeze}} = 5$ post-stop samples. Under sparse communication or high packet-loss conditions, the time to fill these windows grows accordingly, and we revisit this dependency in Section 8.3.

Check 3a (Trajectory consistency). Over a sliding window of $W = 6$ BSMs (0.6 s at 10 Hz, covering the initial braking phase during which position divergence becomes measurable), we compare expected displacement from reported kinematics against actual position change:

$$\delta_j^{(t)} = \left| |\mathbf{x}_j^{(t)} - \mathbf{x}_j^{(t-1)}| - \left(\tilde{v}_j^{(t-1)} \Delta t + \frac{1}{2} \tilde{a}_j^{(t-1)} \Delta t^2 \right) \right|. \quad (8)$$

Let $\tau_x^{(t)} = \max(3.0, 0.20 \cdot \tilde{v}_j^{(t)}) \text{ m}$ be a speed-scaled tolerance, with the base value of 3.0 m accommodating SUMO’s discrete-time position integration artifacts. If more than 50% of intervals in the window satisfy $\delta_j^{(t)} > \tau_x^{(t)}$, the trajectory is flagged ($w_{3a} = 0.20$). The 50% threshold requires persistent inconsistency across at least 4 of 6 BSMs, filtering out transient noise while catching sustained trajectory divergence.

Check 3b (Post-EEBL position-derived speed). This is the primary check for $\mathcal{A}_1$. After an EEBL trigger at t_{trigger} , we monitor the position-derived speed:

$$\hat{v}_j^{(t)} = \frac{|\mathbf{x}_j^{(t)} - \mathbf{x}_j^{(t-1)}|}{\Delta t}, \quad t > t_{\text{trigger}}. \quad (9)$$

If $\hat{v}_j^{(t)} > 0.8 \cdot \hat{v}_j^{(t_{\text{trigger}})}$ for more than 60% of the post-trigger BSMs within the observation window, the vehicle never actually slowed ($w_{3b} = 0.30$). The threshold of 0.8 reflects the expected speed reduction under genuine hard braking: at the EEBL trigger threshold of $0.4g = 3.92 \text{ m/s}^2$ [35], a vehicle traveling at 25 m/s would reduce its speed by 20% within approximately 1.3 s, making this a conservative bound that avoids flagging genuine braking while still detecting the sustained high-speed movement characteristic of $\mathcal{A}_1$. As discussed in Section 4, this check exploits the fact that position-derived speed bypasses reported speed and acceleration fields. However, it does not discriminate when the attacker also controls position evolution ($\mathcal{A}_2$).

Check 3c (Position–speed mismatch). Per-step comparison of position-derived and reported speed:

$$|\hat{v}_j^{(t)} - \tilde{v}_j^{(t)}| > \tau_v = 6.0 \text{ m/s} \implies \text{flag} \quad (w_{3c} = 0.15). \quad (10)$$

Position-derived speed noise is bounded by $2\sigma_x/\Delta t = 2 \times 0.5 / 0.1 = 10 \text{ m/s}$ in the worst case. In SUMO’s discrete-time simulation, consecutive-position differencing exhibits empirical noise of approximately 2 m/s due to sub-step integration artifacts. The tolerance $\tau_v = 6.0 \text{ m/s}$ provides a conservative 3σ margin over this empirical noise, ensuring that simulation dynamics do not trigger false positives while still catching the sustained divergence produced by $\mathcal{A}_1$, where $\hat{v}_{j^*}$ remains near the pre-attack speed ($\approx 25 \text{ m/s}$) while $\tilde{v}_{j^*}$ approaches zero.

5.5 Stage 4: Post-Stop Position Consistency

Stages 1–3 are effective when the attacker’s trajectory is kinematically inconsistent ($\mathcal{A}_1$), but they cannot distinguish $\mathcal{A}_2$ during its braking phase because all reported fields satisfy kinematic equations. However, VASP’s $\mathcal{A}_2$ implementation reveals a post-attack artifact. After the ghost vehicle completes its simulated stop, the attack logic ceases updating the position field. The ghost’s reported position remains frozen at the final stop location across subsequent BSMs, lacking the sub-meter global positioning system (GPS) jitter ($\sigma_x = 0.5 \text{ m}$) that a physically present vehicle would exhibit.

Check 4 (Frozen position). We monitor whether a sender’s position remains unchanged after it claims to have stopped. Let $\epsilon_x = 0.01 \text{ m}$ be a position-change threshold and $N_{\text{freeze}} = 5$ the monitoring window. If more than 80% of position updates within the window show near-zero change:

$$\frac{\#\{t : |\mathbf{x}_j^{(t)} - \mathbf{x}_j^{(t-1)}| < \epsilon_x\}}{N_{\text{freeze}}} > 0.8 \implies \text{flag}. \quad (11)$$

Table 2: Detection check weights.

Check	Description	w_k
1a	TTC plausibility (high TTC)	0.15
1a'	TTC plausibility (receding)	0.35
1b	New sender	0.20
2a	Physical deceleration limit	0.20
2b	Cross-field consistency	0.15
2c	Brake–speed mismatch	0.15
3a	Trajectory consistency	0.20
3b	Post-EEBL position-derived speed	0.30
3c	Position–speed mismatch	0.15
4	Frozen position after stop	0.60

The sender is flagged ($w_4 = 0.60$) if its average reported speed exceeds 0.5 m/s or the brake flag remains set. This additional condition prevents flagging a legitimately parked vehicle. The tight threshold $\epsilon_x = 0.01 \text{ m}$ targets VASP’s attack implementation, where the ghost’s position field freezes entirely after the simulated stop, producing position changes well below the sub-meter GPS jitter ($\sigma_x = 0.5 \text{ m}$) expected from a physically present vehicle. This check does not trigger during the $\mathcal{A}_2$ braking phase (where position updates are kinematically consistent), but activates once the attacker’s trajectory generator stops and the position field freezes. Because Check 4 triggers on every subsequent BSM from the frozen ghost, it provides sustained detection of $\mathcal{A}_2$ after the initial braking transient. The weight $w_4 = 0.60$ exceeds Θ , enabling immediate detection: a confirmed position freeze is a high-confidence indicator of a fabricated trajectory.

5.6 Scoring and Decision

Failed checks contribute to a cumulative anomaly score:

$$S_j^{(t)} = \sum_{k \in \mathcal{K}_{\text{active}}} w_k \cdot \mathbf{1}[\text{check}_k \text{ failed}], \quad (12)$$

where $\mathcal{K}_{\text{active}} \subseteq \{1a, 1a', 1b, 2a, 2b, 2c, 3a, 3b, 3c, 4\}$ is the set of checks applicable at time t , with the constraint that exactly one of $\{1a, 1a'\}$ appears in $\mathcal{K}_{\text{active}}$ for any given BSM (mutually exclusive on the sender’s longitudinal/TTC context). Table 2 lists all weights w_k . Stages 1–2 require the EEBL gate to be open; Stage 3 additionally requires $|\mathcal{H}_j| \geq N_{\text{min}}$ to ensure sufficient trajectory history; Stage 4 requires the sender to have reported a near-zero speed. Since Checks 1a and 1a' are mutually exclusive, the maximum achievable score is 2.30. With the exception of Check 4 ($w_4 = 0.60 > \Theta$), all individual checks have weights below Θ , so detection generally requires corroborating evidence from multiple checks. Check 4 can independently trigger detection: a confirmed position freeze is a high-confidence indicator warranting immediate flagging. A BSM is flagged as malicious if:

$$S_j^{(t)} \geq \Theta = 0.55. \quad (13)$$

We calibrate $\Theta = 0.55$ empirically to balance detection against false-positive cost; sensitivity to this choice is analyzed in Section 7. The individual check weights w_k map to semantic confidence: $w_4 = 0.60$ (confirmed freeze, high confidence), $w_{3b} = 0.30$ (sustained high speed, strong but not conclusive), with the remaining $w \in [0.15, 0.20]$ providing supporting evidence. Tolerance thresholds and window lengths are likewise empirically selected rather than optimized, with the full per-parameter rationale (standards-based thresholds, physics- and noise-derived tolerances, BSM-rate-driven windows) provided in the project repository (docs/parameter_derivation.md).

5.7 IDM Mitigation Layer

When a BSM passes the EEBL gate but is *not* flagged as malicious ($S_j^{(t)} < \Theta$), the ego decelerates according to the IDM-prescribed acceleration rather than executing maximum emergency braking:

$$a_i^{\text{mitigated}} = \max\left(a_{\text{IDM}}\left(v_i^{(t)}, s_{ij}^{(t)}, \Delta v_{ij}^{(t)}\right), -2b\right). \quad (14)$$

The lower bound $-2b = -4.0 \text{ m/s}^2$ caps the IDM output at twice the comfortable deceleration, corresponding to the SAE J2945 hard-brake threshold of $0.4g$ [35]. If the EEBL is flagged ($S_j^{(t)} \geq \Theta$), the ego ignores the warning and continues normal driving. The IDM layer thus serves as a *false-negative safety net*: when a fake EEBL evades detection, the ego’s response is still bounded by IDM rather than an unconstrained emergency brake. If the event is genuine, IDM produces appropriate braking for the current gap and speed [43]. This is especially important for $\mathcal{A}_2$, where detection is inherently limited: even on a missed detection, IDM bounds deceleration to a level consistent with the current gap rather than commanding a full emergency stop.

5.8 Algorithm

Algorithm 1 presents the end-to-end detection and mitigation pipeline. For each incoming BSM, the detector first evaluates the EEBL gate conditions (line 3). BSMs that do not satisfy the gate are discarded without further processing. When the gate is open, Stages 1–4 are executed sequentially, each contributing weighted evidence to the anomaly score $S_j^{(t)}$. If the score exceeds Θ , the sender is flagged and its EEBL warning is suppressed (line 16); otherwise, the ego vehicle responds with IDM-bounded deceleration rather than unconstrained emergency braking (line 18).

Algorithm 1 MIRAGE: Multi-stage Inspection and Response Against Ghost-vehicle EEBL

Require: BSM stream; Θ , T_{warn} , N_{min} , W , N_{freeze}

```

1: for each incoming BSM  $m_j^{(t)}$  from vehicle  $j$  do
2:   Append  $m_j^{(t)}$  to history  $\mathcal{H}_j$ ; set  $S_j^{(t)} \leftarrow 0$ 
3:   if  $\beta_j^{(t)} = 1$  and  $|\mathbf{x}_j^{(t)} - \mathbf{x}_i^{(t)}| \leq R_{\text{max}}$  and  $\text{consecBrake}(j) \geq 2$ 
     then ▷ EEBL Gate
4:     Stage 1: Checks 1a–1b (Eqs. 3–4)
5:     Stage 2: Checks 2a–2c (Eqs. 5–7)
6:     if  $|\mathcal{H}_j| \geq N_{\text{min}}$  then
7:       Stage 3: Checks 3a–3c (Eqs. 8–10)
8:     end if
9:     if  $\hat{v}_j^{(t)} \approx 0$  and position frozen then
10:      Stage 4: Check 4 (Eq. 11)
11:    end if
12:  end if
13:  if  $j$  ahead of  $i$  and  $s_{ij}^{(t)} > 0$  then
14:    Compute  $a_{\text{IDM}}$  via Eq. 14
15:  end if
16:  if  $S_j^{(t)} \geq \Theta$  then
17:    Flag  $j$ ; ignore EEBL from  $j$ 
18:  else if EEBL active:
19:    Respond with  $a_i^{\text{mitigated}} = a_{\text{IDM}}$  (Eq. 14)
20:  end if
21: end for
```

6 Evaluation

6.1 Simulation Platform

We implement MIRAGE within the VASP framework [5], which couples the OMNeT++ discrete-event network simulator with the SUMO traffic simulator via the Veins V2X simulation framework. VASP provides native implementations of 68 BSM attack types, including the two fake EEBL variants evaluated in this work. BSMs are generated at 10 Hz ($\Delta t = 0.1 \text{ s}$) and processed by each vehicle’s on-board detector. This integrated evaluation captures realistic vehicle dynamics and BSM exchange while isolating detection logic from physical-layer effects by running under ideal channel conditions. This is an idealization; packet loss, latency, and congestion effects on the windowed checks are discussed in Section 8.3. By evaluating all detectors within the same framework using VASP’s native attack implementations, we ensure a fair comparison: every method processes identical BSM streams under identical traffic conditions, eliminating confounds from differing attack models or simulation fidelity.

6.2 Scenario Configuration

We simulate a single-lane highway with $N = 20$ SUMO-controlled vehicles over 120 s. Vehicle dynamics are governed by SUMO’s IDM car-following model with driver imperfection $\sigma = 0.5$ and speed deviation $\pm 10\%$, and BSMs

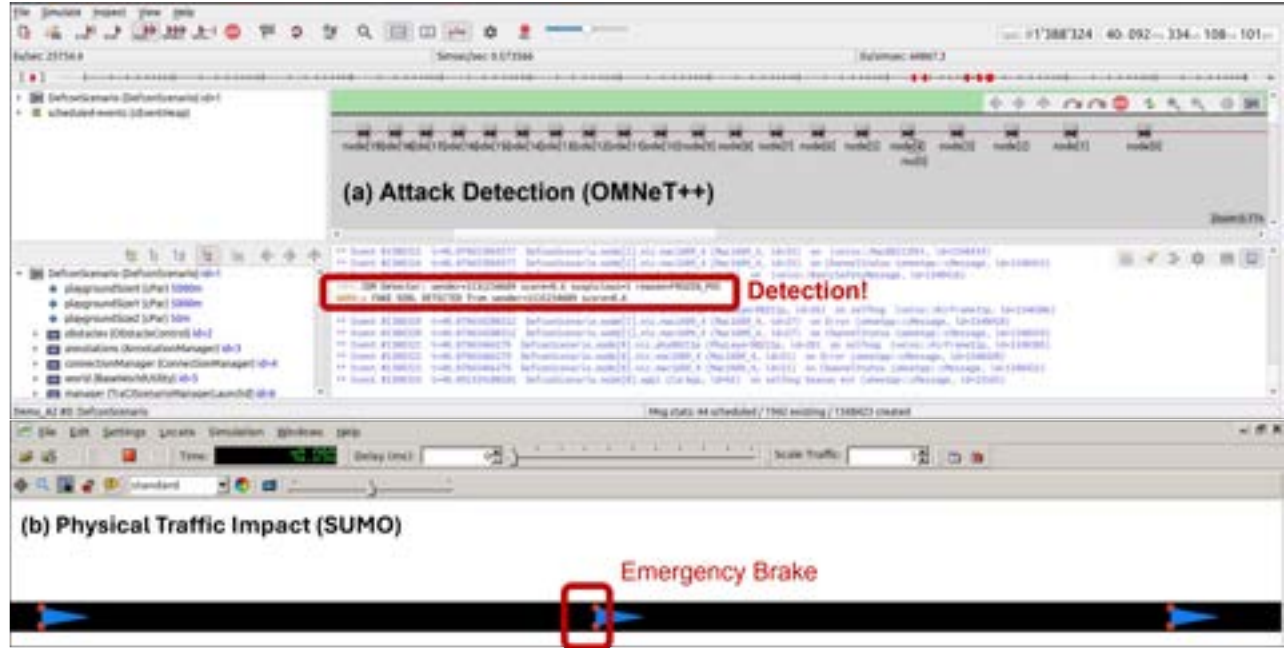


Figure 4: Visualization of the Fake EEBL attack scenario. **(Top)** The OMNeT++ log window shows the detection of a malicious "Fake EEBL" message from the attacker vehicle. **(Bottom)** The SUMO simulation view illustrates the consequence, where trailing vehicles perform emergency braking (highlighted in red), disrupting normal traffic flow.

are exchanged via Veins at 10 Hz. IDM parameters for the mitigation layer follow standard calibrations [23, 43]: $v_0 = 33.33$ m/s, $T = 1.5$ s, $a_{\max} = 1.4$ m/s², $b = 2.0$ m/s², $s_0 = 2.0$ m. To establish statistical reliability, we run each configuration across five seeds with randomized initial vehicle speeds (25 ± 2 m/s) and inter-vehicle gaps (30 ± 3 m), varying both the OMNeT++ random seed and the SUMO route file per run.

One designated vehicle (node 19) acts as the attacker, as shown in Figure 4. In attack scenarios, it executes VASP’s native $\mathcal{A}_1$ or $\mathcal{A}_2$ implementation, injecting a ghost vehicle that broadcasts fabricated EEBL BSMs to all vehicles within communication range. Each of the remaining 19 vehicles independently evaluates incoming BSMs using its on-board detector. We run separate simulations for baseline (no attack), $\mathcal{A}_1$, and $\mathcal{A}_2$ for each detector configuration.

6.3 Baselines

We compare against three baselines spanning the range of existing MDS approaches. Table 3 summarizes the differences. We use each baseline’s published default thresholds and configuration; tuning baselines specifically for the fake-EEBL setting could shift their per-attack tradeoffs but is unlikely to remove the cross-attack imbalance documented in RQ2 (Threshold and F2MD detect $\mathcal{A}_1$ -style decelerations but miss $\mathcal{A}_2$, while VCADS exhibits the opposite pattern). Event-gated variants of these baselines were not evaluated and are an in-

Table 3: Comparison of detection approaches.

	B1 [24]	B2 [3]	B3 [5, 20]	MIRAGE
Activation	Always	Always	Always	Event-gated
Check type	Threshold	Cross-field	Multi-check	Multi-stage
Post-event traj.	×	×	×	✓
Post-stop check	×	×	×	✓
Weighted scoring	×	×	×	✓
Mitigation	×	×	×	IDM-based

teresting direction for follow-up work.

B1: Threshold [24]. Range-and-threshold plausibility checks [6, 24] that evaluate individual BSM fields against fixed bounds ($\tilde{v}_j < 70$ m/s, $|\tilde{a}_j| < 11$ m/s², $s_{ij} < 300$ m), augmented with a basic brake-deceleration consistency rule that compares the brake flag against reported deceleration.

B2: VCADS [3]. Physics-based cross-field validation using a sliding window of $W = 5$ BSMs. VCADS computes three cross-checks between reported and derived kinematics: acceleration, position-displacement, and speed consistencies. Although it uses a temporal window, all checks operate on instantaneous field pairs within the window rather than analyzing post-event trajectory patterns.

B3: F2MD/VASP [5, 20]. A multi-check plausibility chain combining six checks: communication and speed range verification, position jump rate, BSM frequency, and position-speed/acceleration consistency. The checks span both per-

field thresholds and cross-field comparisons but are applied uniformly to every BSM without application-specific gating.

6.4 Metrics

We evaluate detection performance using four metrics at the per-BSM granularity: true positive rate (TPR), false positive rate (FPR), precision, and F1 score. A true positive is an attack BSM correctly flagged by the detector, and a false positive is a benign BSM incorrectly flagged.

In the EEBL context, FPR carries asymmetric cost: a false positive suppresses a genuine emergency warning, whereas a false negative results in unnecessary but recoverable braking [45]. We also report per-attack breakdowns to separate the performance of $\mathcal{A}_1$ and $\mathcal{A}_2$, since aggregate metrics can mask important asymmetries.

7 Results

Each detector processes approximately 4.7 million BSM evaluations per seed (3.7M benign, 1.0M attack) across five seeds with randomized traffic conditions ($\sigma=0.5$, varied initial speeds and gaps). We evaluate MIRAGE to answer the following research questions (RQ):

RQ1 (Effectiveness of MIRAGE): How accurately does MIRAGE identify fake EEBL messages?

RQ2 (Baseline Comparison): How does MIRAGE perform relative to existing detectors?

RQ3 (Per-Attack Analysis): How does detection performance differ between the two attack variants $\mathcal{A}_1$ and $\mathcal{A}_2$?

RQ4 (Parameter Sensitivity): How robust is MIRAGE’s detection accuracy to variations in the decision threshold Θ ?

RQ5 (Response Mitigation): How much does the IDM-based mitigation limit braking severity on undetected attack BSMs?

7.1 RQ1: Effectiveness of MIRAGE

Table 4 summarizes aggregate performance across five seeds. MIRAGE achieves an F1 score of 0.922 ± 0.001 at the operating point $\Theta = 0.55$, with TPR = 0.883, FPR = 0.009, and precision = 0.965. The low FPR is particularly noteworthy in the EEBL context: Van der Heijden et al. [45] emphasize that false positives in safety applications carry asymmetric cost, as suppressing a genuine emergency warning may directly cause a collision. Several baselines also achieve low per-BSM FPR (Threshold and F2MD both at 0.009; see Table 4); the distinguishing property of MIRAGE is that it achieves this low FPR while simultaneously maintaining high TPR across both attack variants, where Threshold and F2MD trade FPR for substantial recall loss on at least one variant. Event-gating contributes to this profile by restricting deep inspection to EEBL-active periods, which keeps false alarm pressure low even as the detector applies more sensitive multi-stage checks. The balanced per-attack coverage (77.0% for $\mathcal{A}_1$, 99.7% for

Table 4: Aggregate detection performance averaged over 5 seeds. Standard deviations in parentheses.

Method	TPR	FPR	Precision	F1 score
B1: Threshold	0.597 (0.003)	0.009 (0.001)	0.950 (0.009)	0.733 (0.004)
B2: VCADS	0.499 (0.000)	0.040 (0.011)	0.778 (0.059)	0.607 (0.018)
B3: F2MD/VASP	0.110 (0.003)	0.009 (0.001)	0.770 (0.037)	0.193 (0.006)
MIRAGE	0.883 (0.003)	0.009 (0.001)	0.965 (0.007)	0.922 (0.001)

$\mathcal{A}_2$ on VASP’s current $\mathcal{A}_2$ implementation; see RQ3) confirms that the aggregate F1 reflects consistent detection rather than strength on a single variant.

7.2 RQ2: Baseline Comparison

Table 4 compares MIRAGE against three baselines. MIRAGE substantially outperforms VCADS (F1 = 0.607), Threshold (F1 = 0.733), and F2MD/VASP (F1 = 0.193). Threshold achieves moderate TPR (0.597) by detecting primarily $\mathcal{A}_2$ through its brake-deceleration consistency, but misses most $\mathcal{A}_1$ instances. VCADS detects $\mathcal{A}_1$ effectively through cross-field validation but fails on the kinematically consistent $\mathcal{A}_2$, yielding low aggregate TPR (0.499) despite high per-attack accuracy on $\mathcal{A}_1$. F2MD/VASP achieves the lowest detection rate (TPR = 0.110) because it was designed for VeReMi-style misbehavior [21, 46], where attacks produce extreme BSM field values. Fake EEBL attacks create kinematically plausible values that pass its checks. MIRAGE’s advantage stems from combining trajectory-based checks (effective against $\mathcal{A}_1$) with post-stop position consistency (effective against $\mathcal{A}_2$), a complementarity that no single baseline achieves.

7.3 RQ3: Per-Attack Detection Analysis

Figure 5 separates results by attack type. Each baseline detector shows a pronounced specialization toward one attack variant, whereas MIRAGE is the only detector to maintain high TPR across both.

For $\mathcal{A}_1$ (NoStop), VCADS achieves the highest per-attack TPR (0.997) through its cross-field kinematic validation, which detects the speed-position divergence inherent in this attack. MIRAGE reaches TPR = 0.770, driven primarily by Check 3b (post-EEBL position-derived speed). Threshold and F2MD both achieve approximately 0.22, as $\mathcal{A}_1$ ’s fabricated values remain within their plausibility bounds.

For $\mathcal{A}_2$ (WithStop), the situation reverses. VCADS achieves TPR = 0.000 because the ghost’s fields are mutually consistent during the braking phase, leaving no cross-field discrepancy to detect. Threshold detects $\mathcal{A}_2$ at 0.976 through its brake-

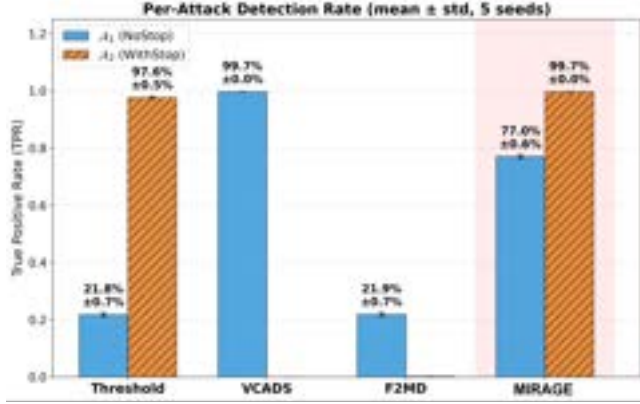


Figure 5: Per-attack detection rate (TPR). MIRAGE is the only detector that maintains high TPR across both $\mathcal{A}_1$ and $\mathcal{A}_2$.

deceleration consistency rule in the post-stop phase. After the ghost completes its simulated stop, the brake flag remains set while the speed-derived deceleration drops to zero, triggering the brake-without-deceleration check. This detection, similar to our Stage 4, depends on post-stop behavior rather than the kinematically consistent braking phase. MIRAGE achieves TPR = 0.997, with detection driven by Stage 4: after the ghost completes its simulated stop, its position field freezes, and Check 4 flags the absence of expected GPS jitter.

This complementarity is the core result. Existing detectors are each effective against one attack variant but fail against the other, whereas MIRAGE’s multi-stage architecture provides balanced coverage. The F1 score of 0.922 reflects this balanced performance rather than strength in a single variant.

We note that MIRAGE’s $\mathcal{A}_1$ TPR (0.770) is lower than VCADS (0.997) because our position-derived speed check (Check 3b) requires a short observation window before the divergence becomes statistically distinguishable from sensor noise. During the first few BSMs after the EEBL trigger, the position displacement difference is within the noise margin, and these early BSMs are scored below Θ . This reflects a deliberate design choice: we prioritize low FPR over maximizing per-attack TPR for $\mathcal{A}_1$, since our IDM mitigation layer bounds the ego’s response even for undetected attack BSMs during this transient window. In contrast, VCADS achieves near-perfect $\mathcal{A}_1$ detection but with an FPR of 0.040, more than four times ours (0.009), and zero detection of $\mathcal{A}_2$.

7.4 RQ4: Parameter Sensitivity

Figure 6 shows TPR, FPR, and F1 as functions of Θ (mean ± std across five seeds). The best F1 (0.925) occurs at $\Theta = 0.52$, close to our chosen operating point of 0.55. Performance is stable across $\Theta \in [0.45, 0.59]$: TPR remains above 0.88 and F1 above 0.92, indicating that the detector is robust to the

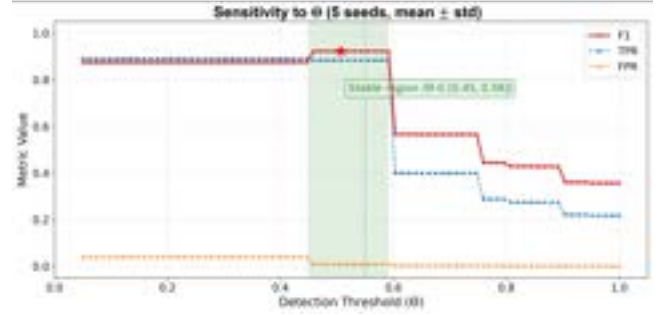


Figure 6: Detection performance as a function of Θ . The shaded region marks the stable operating range. The star marks the best F1 score.

exact threshold choice.

Below $\Theta = 0.45$, FPR increases to approximately 0.05 as lower-confidence check combinations begin to trigger false alarms. At $\Theta = 0.60$, TPR drops from 0.883 to 0.41 and F1 drops to 0.57 because Check 4’s weight ($w_4 = 0.60$) no longer exceeds the threshold, collapsing $\mathcal{A}_2$ detection entirely. Above $\Theta = 0.75$, TPR falls below 0.30 as even multi-check attack evidence falls below the threshold. We selected $\Theta = 0.55$ as it lies within the stable region and slightly favors detection coverage over the F1-optimal 0.52. The marginal FPR difference between 0.52 and 0.55 is acceptable in our deployment model. MIRAGE is intended as a complement to PKI-level revocation rather than a sole decision authority. False positives produce ignored EEBL warnings rather than safety actions, and the IDM mitigation layer is invoked when detection *misses* an attack, not when detection produces a false alarm.

7.5 RQ5: Response Mitigation

Table 5 summarizes the IDM-bounded deceleration for EEBL-relevant BSMs. For attack BSMs, the IDM-bounded deceleration averages -2.03 m/s^2 (median -1.86 m/s^2), with a maximum magnitude of -4.0 m/s^2 . This compares to the -8.0 m/s^2 ($\approx 0.8g$) maximum emergency braking that an unprotected EEBL response would command, representing a 75% reduction in mean braking severity and a 50% reduction in peak deceleration magnitude. For benign BSMs in EEBL-relevant contexts, IDM output averages -1.75 m/s^2 (median -1.36 m/s^2), consistent with normal car-following deceleration. Both distributions are bounded well above the emergency braking threshold, confirming that IDM limits the ego’s response to physics-reasonable levels regardless of whether the triggering BSM is genuine or fabricated.

The IDM layer provides defense in depth: for the small fraction of attack BSMs that evade detection (primarily during $\mathcal{A}_2$ ’s initial braking phase before Stage 4 activates), the ego’s response is bounded to a level consistent with the observed

Table 5: IDM-bounded deceleration for EEBL BSMs.

BSM Type	Mean	Median	Min
Attack ($n=194,100$)	-2.03 m/s^2	-1.86 m/s^2	-4.00 m/s^2
Benign ($n=1,292,323$)	-1.75 m/s^2	-1.36 m/s^2	-4.00 m/s^2
Unprotected EEBL	-8.0 m/s^2 (emergency braking)		

gap and closing speed rather than an unconstrained emergency stop. This fail-safe property is particularly valuable given the safety-critical nature of EEBL, where false negatives directly translate to unnecessary hard braking.

To contextualize this reduction, the IDM-bounded peak of -4.0 m/s^2 ($\approx 0.4g$) corresponds to the SAE J2945 hard-brake threshold itself [35], meaning that even in the worst case, the ego’s response does not exceed the deceleration level that would normally trigger an EEBL warning. By contrast, an unprotected response at -8.0 m/s^2 ($\approx 0.8g$) approaches the physical braking limit of production vehicles [13], substantially increasing rear-end collision risk. This quantitative bound distinguishes our approach from binary accept/reject detectors, which provide no mitigation on missed detections.

8 Discussion

We discuss MIRAGE’s detection behavior on the evaluated attacks (Section 8.1), its deployment considerations and generalization potential beyond EEBL (Section 8.2), and the explicit limitations that scope its claims (Section 8.3).

8.1 Detection Analysis

Detecting $\mathcal{A}_2$ via Post-Attack Artifacts. FakeEEBL_WithStop produces kinematically consistent BSMs during its braking phase, making it indistinguishable from a valid hard-brake trajectory at the field level, a well-known evasion principle [5, 32]. Our Stage 4 check instead targets the ghost’s post-attack trajectory artifact: in VASP’s implementation, the position field freezes after the simulated stop, producing changes below $\epsilon_x = 0.01 \text{ m}$ across consecutive BSMs. Check 4 detects this with 99.7% TPR and carries the highest weight ($w_4 = 0.60 > \Theta$), enabling immediate flagging once the freeze is confirmed. An adaptive attacker could instead emit plausible post-stop updates (see Adaptive Attacker Analysis below); the IDM mitigation serves as the safety net in that case.

Per-Check Contribution Analysis. An ablation shows that Check 4 (position freeze, $w=0.60$) drives $\mathcal{A}_2$ detection almost exclusively ($\Delta\text{TPR} = -0.974$ when removed), while Check 3b (post-EEBL no stop, $w=0.30$) is the primary contributor for $\mathcal{A}_1$ ($\Delta\text{TPR} = -0.222$). The remaining checks have a marginal impact against the two evaluated variants, since these attacks do not trigger the anomalies those checks target

(physical-limit violations, cross-field inconsistencies); we retain them as defense-in-depth layers for more adaptive future attacks. The full ablation figure (ΔF1 and ΔTPR across Overall, $\mathcal{A}_1$, $\mathcal{A}_2$ for each removed check) is provided in the project repository (docs/extended_discussion.md).

Adaptive Attacker Analysis. Defenses validated against fixed attack implementations may fail against adaptive adversaries [5, 32], so we make MIRAGE’s threat envelope explicit through three adaptive-attacker classes beyond $\mathcal{A}_1$, $\mathcal{A}_2$. *Path 1 (GPS noise)* injects jitter above $\epsilon_x = 0.01 \text{ m}$ to defeat Check 4, but no fixed threshold resolves this (values below the genuine jitter envelope $\sigma_x \approx 0.5 \text{ m}$ are replicable; above it causes false positives), so discrimination needs evidence beyond a single sender’s stream [3, 39]. *Path 2 (continued plausible updates)* never triggers Check 4 and is the practical upper bound for content-only detection, though Checks 3b and 3c ($\tau_v = 6.0 \text{ m/s}$) still narrow the evasion. *Path 3 (history-building, pseudonym change)* does not directly evade MIRAGE but maps to the SCMS/revocation layer, out of scope [34, 49]. Because content-only detection cannot close Path 2, the IDM layer caps ego deceleration at -4.0 m/s^2 regardless of detection [4, 37], reducing worst-case braking from -8.0 m/s^2 to the SAE J2945 threshold [35] as a mitigation, not a guarantee. None of these undermine the event-gating design; a per-path breakdown is in the project repository (docs/extended_discussion.md).

8.2 Deployment Considerations

Safety vs. Security Tradeoff. Suppressing a genuine EEBL is potentially more dangerous than acting on a fake one [45]. Our design reflects this asymmetry: $\Theta = 0.55$ tolerates some missed attacks while keeping a low FPR, and IDM mitigation ensures graceful degradation rather than binary accept/reject, aligning with fail-safe recommendations in V2X standardization [4, 37].

Authentication Without Authorization. The vehicular PKI and SCMS [49] authenticate that a BSM originates from an enrolled device but impose no authorization on message content, so any enrolled OBU may broadcast any kinematic values. Proposed content-level validation mechanisms (sensor-aided cross-checking [3], cooperative witness protocols [4], reputation systems [10, 12]) are not yet adopted in deployment standards [8, 35]. MIRAGE operates within this gap, requiring no infrastructure changes, extra message exchanges, or SCMS modifications, and performing all checks locally on standard BSM fields.

Generalization Beyond EEBL. The event-gated architecture is a design pattern, not a result specific to EEBL: it requires a well-defined trigger condition, application-specific behavioral invariants a genuine event must satisfy, and a controller-level layer that bounds the ego’s reaction under uncertain detection. Other longitudinal V2X safety applications (e.g., Forward

Collision Warning) fit the same template, whereas lateral applications (Intersection Movement Assist, Lane Change Warning) would require different invariants and steering-rate or lateral-acceleration bounds rather than the IDM deceleration cap. The transferable elements (gating principle, windowed-evidence aggregation, weighted scoring) and the application-specific ones that must be re-derived per standard (trigger conditions, invariants, tolerances, weights) are detailed in the project repository (`docs/extended_discussion.md`). We therefore present event-gating as a transferable design pattern while restricting empirical claims strictly to the evaluated EEBL variants ($\mathcal{A}_1$, $\mathcal{A}_2$).

Scalability in Dense Traffic. MIRAGE’s per-BSM checks are closed-form arithmetic, so per-sender state grows linearly with the number of neighbors N and processing remains $O(1)$ per message ($O(N)$ per second at 10 Hz). Full Stage 1–4 evaluation runs only when the EEBL gate opens, so the full-pipeline rate is bounded by the (sparse) rate of genuine hard-brake events rather than the total BSM rate. Channel congestion and packet loss in dense traffic degrade the sliding-window checks (Section 8.3, item 3) rather than the algorithmic scalability, and each sender is evaluated independently. A detailed treatment of all four scaling concerns is provided in the project repository (`docs/extended_discussion.md`); quantifying performance under dense, lossy-channel operation is left for future work.

Simulation Realism and Computational Overhead. A potential concern with simulation-based evaluation is whether the results transfer to real-world V2X deployments. We address this through three design choices. First, SUMO’s car-following models are extensively calibrated against real-world trajectory data [23, 26], and the IDM parameters we adopt ($v_0 = 33.33$ m/s, $T = 1.5$ s, $b = 2.0$ m/s²) follow standard calibrations that reproduce empirical speed-flow relationships [43]. Second, the sensor noise parameters ($\sigma_x = 0.5$ m, $\sigma_a = 0.15$ m/s²) reflect typical GPS and IMU error bounds reported in V2X field operational tests [18, 22]. Third, evaluating all methods on VASP’s native attack implementations ensures the comparison is not biased by our own attack modeling. Ideal channel conditions represent a best-case assumption; the impact of packet loss and channel congestion is discussed under Section 8.3.

All detection checks are closed-form arithmetic (comparisons, subtractions, divisions) with no iterative solvers, cryptographic operations, or ML inference, and the full pipeline runs only when the EEBL gate is open; during normal driving, incoming BSMs are dismissed by the gate test with no further processing. In our evaluation, each of the 19 non-attacker vehicles ran the full detector independently in OMNeT++, processing about 4.7 million BSM evaluations per seed without simulation delays or relaxation of the 10 Hz rate, confirming that per-BSM processing fits within the $\Delta t = 0.1$ s BSM interval. MIRAGE flags attacks within 0.14 s for $\mathcal{A}_1$ and 0.29 s for $\mathcal{A}_2$ (mean, across the 19 ego vehicles over 5

seeds), faster than all baselines: VCADS detects $\mathcal{A}_1$ comparably (0.24 s) but fails entirely on $\mathcal{A}_2$, while Threshold and F2MD require 1.5 to 9.7 s on average. Even during the pre-detection window, the IDM mitigation layer caps the ego’s deceleration at -4.0 m/s². The full end-to-end detection-latency breakdown (mean, median, and missed-BSM counts per detector and attack variant) is provided in the project repository (`docs/extended_discussion.md`).

8.3 Limitations

MIRAGE is an application-specific detector evaluated on VASP’s two fake-EEBL implementations under single-lane, ideal-channel simulation. We make explicit what we do and do not claim. *We claim:* the event-gated architecture is effective against the two VASP-implemented fake-EEBL variants ($\mathcal{A}_1$, $\mathcal{A}_2$) under the evaluated conditions; the IDM bounding layer reduces worst-case ego deceleration even when detection misses. *We do not claim:* robustness against adaptive trajectory generators that defeat Check 4; generalization to multi-lane or urban traffic; performance under lossy or congested wireless channels; generalization to V2X services beyond EEBL; or optimality of the chosen weights and thresholds. The six items below scope each of these areas.

1. Artifact dependency and adaptive attackers. The 99.7% detection rate on $\mathcal{A}_2$ is driven by a specific implementation artifact in VASP’s FakeEEBL_WithStop generator: the reported position field freezes at the claimed stop location. This headline number should not be taken as reliable evidence of physically plausible $\mathcal{A}_2$ instances; an adaptive attacker that injects synthetic GPS noise or continues plausible position updates can defeat Check 4 (Adaptive Attacker Analysis, Paths 1 and 2 in Section 8.1). Closing this gap requires sensor-aided cross-validation [3], cooperative corroboration [4], or physical-layer features [39]; the IDM layer complements these as a safety net rather than as a defense.

2. Single-lane evaluation. Our evaluation is restricted to a single-lane highway. Multi-lane traffic introduces three effects: lateral position ambiguity (adjacent-lane braking can open the EEBL gate, likely raising the false-positive rate), lane-association errors during lane changes (genuine post-braking trajectories may appear inconsistent with Check 3a), and sender- vs. ego-side lane-change transients (perturbing Check 3a, TTC, and relevance gating). Multi-lane extension likely requires map-aware lane association.

3. Ideal channel assumption. Our simulations assume ideal V2X channels. Deployed communication can be lossy (multipath fading, NLOS shadowing, collisions on the shared 5.9 GHz spectrum, congestion) [20]: packet loss reduces the BSM history available to Stages 3 and 4, delaying window fills and potentially raising false negatives for short attacks, while latency and out-of-order arrival degrade Check 3c’s position-derived speed estimate. Quantifying these effects us-

ing realistic channel models or hardware-in-the-loop testbeds is left for future work.

4. BSM history requirement. Stage 3 requires $N_{\min} = 3$ BSMs per sender to activate. Under high packet loss, window-fill times increase. Senders that appear briefly and then disappear (e.g., pseudonym-change attackers) may not accumulate sufficient history; addressing this requires SCMS-level integration [49] outside our content-level scope.

5. Parameter calibration. Weights, tolerances, and thresholds are empirically selected rather than jointly optimized, and the calibration is valid for the evaluated single-lane highway scenarios with ideal channels. Each parameter has a principled rationale (physical limits, sensor noise envelopes, BSM rate, braking dynamics), detailed in the project repository (`docs/parameter_derivation.md`), and we anticipate that some thresholds (notably τ_v , τ_r , and the speed-scaled term in Check 3a) require recalibration for multi-lane or lossy-channel deployment. Systematic joint sensitivity analysis or learning-based optimization [15, 38] could improve robustness and remains an important direction for future work.

6. Scope of empirical claims. Empirical results come from a single simulator (VEINS 5.2 + SUMO + VASP), a single highway scenario, and a single attack family (fake EEBL); we have not evaluated MIRAGE on hardware testbeds, urban/intersection scenarios, or other V2X services, and coordinated multi-attacker, Sybil, and pseudonym-chain attacks [34, 49] are out of scope. Metrics are reported per-BSM, which can understate the safety impact (event-level safety depends on time-to-first-flag; see `docs/extended_discussion.md`), and the IDM mitigation bounds deceleration rather than guaranteeing collision avoidance. Specific numbers should not be extrapolated beyond this envelope without re-evaluation.

9 Related Work

V2X Data Attacks and EEBL. False safety messages have been studied since early VANET research [14, 19, 33]. Lo and Tsai [25] formalized illusion attacks, and Petit and Shladover [31] cataloged cyberattacks on connected vehicles. Amoozadeh et al. [2] showed how falsified messages cause string instability in platoons. VeReMi [46] and its extension [21] provide labeled misbehavior datasets, while VASP [5] implements 68 BSM attacks, including two targeting EEBL. Beyond the vehicular domain, systematic attack discovery frameworks have also been developed for adjacent autonomous-agent systems, such as Raven [51], which uses signal temporal logic and stochastic optimization to discover semantic false-data injection attacks against multi-robot collision-avoidance algorithms. Despite these advances, detection work specifically addressing fake EEBL remains sparse.

Misbehavior Detection Systems. Plausibility-based detection compares BSM fields against physical bounds [6, 24, 36]. Stübing et al. [41] proposed path prediction for two-stage Car-

to-X verification. VCADS [3] uses kinematic equations for cross-field validation, and F2MD [20] chains multiple plausibility checks. ML approaches include support vector machine and neural network classifiers [15], consecutive BSM feature analysis [38], and RSSI-based physical-layer checks [39]. Recent surveys [7, 37, 45] classify these into data-centric and node-centric categories.

Our work differs from existing MDS not by introducing individual components but by integrating application-specific event gating, trajectory-level behavioral analysis, and physics-grounded response bounding into a single EEBL-specific defense. Three aspects of this integration are specific to our work. First, we validate the EEBL *event* rather than individual BSM fields; this application-specific gating reduces the false-alarm surface that plagues always-on detectors in safety-critical contexts [45]. Second, we check *post-event behavioral consistency* through position-derived speed, a quantity that existing cross-field validators do not exploit because they operate on instantaneous BSM pairs rather than post-trigger trajectory windows. Third, we incorporate an IDM-based mitigation layer that bounds the ego’s physical response, providing a fail-safe mechanism absent from all prior detection-only approaches [7, 37]. To our knowledge, this integration of application-specific event gating, trajectory-level behavioral analysis, and physics-grounded response bounding has not been explored in the V2X misbehavior detection literature.

Resilient V2X Response. Beyond detection, the literature has explored credential revocation [34, 49], reputation-based trust [10, 12], blockchain-based consensus [50], and sensor-aided validation [3, 39]. In platooning, bounded filtering limits the impact of malicious commands [2]. Our IDM-based mitigation extends this principle to ad hoc traffic by bounding the ego’s response using a car-following model [43]. To our knowledge, this integration of detection and response bounding within a single EEBL-specific framework has not been previously proposed.

10 Conclusion

We presented MIRAGE, an event-gated multi-stage misbehavior detector for fake EEBL attacks grounded in the Intelligent Driver Model. The detector restricts inspection to EEBL-active contexts, validates post-event trajectory consistency through position-derived speed, and detects post-stop position-freezing artifacts. Across five randomized seeds, it achieves an F_1 of 0.922 ± 0.001 at 0.9% FPR, with 77.0% TPR on $\mathcal{A}_1$ and 99.7% on $\mathcal{A}_2$ (the latter driven by an implementation-specific position-freezing artifact in VASP, discussed in Section 8.3), making it the only evaluated detector that maintains high detection across both kinematically inconsistent and self-consistent attacks. The IDM mitigation layer further reduces mean braking severity by 75% and limits peak deceleration to -4.0 m/s^2 for undetected attack BSMs.

Our results confirm that application-specific, event-driven detection outperforms general-purpose BSM plausibility checks for EEBL and suggest that the V2X security community should move toward application-aware detection architectures that exploit the semantics and temporal structure of each safety application. Defending against adaptive trajectory generators that continue to produce plausible position updates remains an open problem. Future work will evaluate the detector under wireless channel impairments, extend event-gating to other V2X services such as Intersection Movement Assist and Forward Collision Warning, explore learning-based weight optimization using larger misbehavior datasets [7, 21], and pursue staged hardware validation from closed-track V2X testbeds to instrumented-vehicle deployments.

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Ethical Considerations

This work studies fake EEBL attacks exclusively in a closed-loop simulation environment (VASP/VEINS/SUMO) with no real vehicles, real roadways, or human participants involved at any stage of the research. All experiments were conducted on synthetic traffic scenarios generated by the SUMO microscopic traffic simulator, and no data from real drivers or production vehicle systems were collected or used.

The two attack variants evaluated ($\mathcal{A}_1$ and $\mathcal{A}_2$) are implemented using VASP's existing native attack modules, which are publicly available as open-source software to the research community [5]. No new attack tools, exploits, or vulnerability disclosures were developed as part of this work; the attack descriptions in Section 3 are limited to what is necessary to explain the detection methodology and reproduce the experimental evaluation. The primary goal of this research is defensive: to improve the detection and mitigation of safety-critical V2X threats that have been identified as high risk in prior threat assessments [5, 52]. We believe MIRAGE and comparative evaluations against existing baselines, substantially outweigh the marginal risk of describing attack vectors that are already thoroughly documented in prior work [25, 32, 52].

Availability

To support reproducibility and independent re-evaluation, the complete MIRAGE artifact (detector source code, automated install pipeline, SUMO/Veins scenarios, analysis scripts, and reference results) is permanently archived on Zenodo at <https://doi.org/10.5281/zenodo.19639252> (concept DOI resolving to the latest version; camera-ready snapshot is version 2.1, <https://doi.org/10.5281/zenodo.20398530>), released under the MIT license, and awarded the Artifacts Available, Functional, and Reproduced badges. Supplementary documentation cited in this paper (in docs/) is maintained in the project repository at <https://github.com/eunhanka/mirage-eebl-detector>.

Use of Generative AI

In accordance with the VehicleSec 2026 policy on the use of generative AI, we disclose that ChatGPT (OpenAI) was used during manuscript preparation for grammar checking and L^AT_EX formatting assistance. No generative AI tool was used to generate technical content, design the detection methodology, implement the detector, conduct experiments, analyze data, or interpret results. All scientific contributions are entirely the work of the authors. The authors have verified all content and assume full responsibility for its originality.

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A Artifact Appendix

A.1 Abstract

This artifact contains the source code, install pipeline, experimental scenarios, reference results, and automated claim verifier for MIRAGE, an event-gated misbehavior detector for fake Emergency Electronic Brake Light (EEBL) attacks in Vehicle-to-Everything (V2X) networks. From a fresh Ubuntu 20.04 host, a single command installs all dependencies (OMNeT++ 5.6.2, SUMO 1.8.0, Veins 5.2, VASP), compiles MIRAGE, and produces a ready-to-run environment; a smoke test, single-seed reproduction, and an automated verifier cover the path from installation to claim validation in under two hours on commodity hardware.

It reproduces the paper’s five quantitative claims: aggregate effectiveness ($F_1 = 0.922$, TPR = 0.883, FPR = 0.009, Precision = 0.965), balanced per-attack coverage (77.0 % on $\mathcal{A}_1$, 99.7 % on $\mathcal{A}_2$), superiority over three baselines (Threshold, VCADS, F2MD), threshold robustness across $\Theta \in [0.45, 0.59]$, and IDM mitigation reducing mean braking severity by approximately 75 %. It was verified end-to-end in three clean-room runs on a fresh ubuntu:20.04 container, all producing 4/4 PASS within 1 h 55 min.

A.2 Description & Requirements

A.2.1 Security, privacy, and ethical concerns

The artifact is purely simulation-based: it does not interact with any real vehicle, live network, or physical radio, does

not disable any host security mechanism, and writes only under /opt/mirage and the scenario directory. Reviewers preferring stricter isolation can run it inside an ubuntu:20.04 Docker container, as the authors did for clean-room verification.

A.2.2 How to access

- **Archival DOI (recommended):** <https://doi.org/10.5281/zenodo.19639252> (Zenodo concept DOI; always resolves to the latest published version).
- **Version DOI of this submission:** <https://doi.org/10.5281/zenodo.20398530> (immutable snapshot, version 2.1). The archived zip has MD5 a0633472ad2c33a39fa4a3914dadca99 and size approximately 6.7 MB. It contains the source code, reference figures and tables, and the reference verifier outputs; raw simulation logs are *not* included and are not required, as every claim is reproducible from a fresh install.
- **GitHub (working copy, for AE phase only):** <https://github.com/eunhanka/mirage-eebl-detector>, branch artifact-evaluation, tag vehiclesec26-ae-v2.1 (commit 2981dd0).
- **License:** MIT, consistent with the upstream VASP project.

A.2.3 Hardware dependencies

Commodity x86_64 workstation; minimum 4 CPU cores, 8 GB RAM, 20 GB free disk; recommended 16 cores, 16 GB RAM. No GPU, FPGA, or specialized hardware is required. The artifact was verified on a 16-core Ubuntu 20.04 host.

A.2.4 Software dependencies

Host OS: **Ubuntu 20.04 LTS**. All dependencies are pinned and installed automatically by scripts/install.sh: GCC 9.4.0 (Ubuntu 20.04 default); OMNeT++ 5.6.2 (upstream tarball); SUMO 1.8.0 (git tag v1_8_0); Veins 5.2 (tag veins-5.2, commit c5b4d7c4); VASP (git commit 0ec4af32); Python 3.8 with numpy, pandas, matplotlib, scikit-learn (via pip); Qt5, Boost, Xerces-C (via apt). Network access is required during installation for apt, pip, and git against github.com; air-gapped evaluation is not supported without prior source staging.

A.2.5 Benchmarks

The evaluation uses the synthetic scenario of Section 7 of the paper: a single-lane highway with 20 SUMO-controlled vehicles over 120 s, governed by the SUMO IDM model with driver imperfection $\sigma = 0.5$, speed deviation $\pm 10\%$,

initial speeds 25 ± 2 m/s, and inter-vehicle gaps 30 ± 3 m. Five random seeds $\{0, 1, 2, 3, 4\}$ are used for the aggregate results. Per-seed traffic variations are generated at runtime by `generate_route_variants.py`, which produces `highway_seed{0..4}.rou.xml` inside the scenario directory. `run_single_seed.sh` invokes this generator automatically on first use; reviewers running `run_multi_seed.sh` as a standalone entry point must invoke it once beforehand (see experiment E1). Attacker behavior is the unmodified VASP implementation of `FakeEEBLJustAttack` ($\mathcal{A}_1$, NoStop) and `FakeEEBLStopPositionUpdateAfterAttack` ($\mathcal{A}_2$, WithStop). No external dataset is required.

A.3 Set-up

From a fresh Ubuntu 20.04 host, cloning the artifact-evaluation branch and running `sudo ./scripts/install.sh` all builds the full pinned stack through five idempotent stages (approximately 7 min on 16 cores, 25 to 30 min on 4 cores; any stage is re-runnable). A smoke test, `./scripts/quick_test.sh`, then runs MIRAGE against $\mathcal{A}_2$ for 120 s and, on success, reports 327868 total and 96990 suspicious rows and exits 0. Exact commands, expected console output, and troubleshooting are in the repository `README.md`.

A.4 Evaluation workflow

A.4.1 Major Claims

- (C1): MIRAGE achieves $F_1 = 0.922$ at $\Theta = 0.55$ (TPR = 0.883, FPR = 0.009, Precision = 0.965), averaged over five seeds (per-metric SD below 0.01; tolerance bands in `CLAIMS.md`). Proven by (E1); paper Table 4.
- (C2): MIRAGE is the only detector sustaining high TPR on both variants (0.770 on $\mathcal{A}_1$, 0.997 on $\mathcal{A}_2$); no baseline exceeds TPR = 0.25 on both. Proven by (E1); paper Figure 5.
- (C3): MIRAGE outperforms three baselines: Threshold ($F_1 = 0.733$), VCADS (0.607), F2MD/VASP (0.193). Proven by (E1); paper Table 4.
- (C4): Detection is stable across $\Theta \in [0.45, 0.59]$ ($F_1 > 0.92$), with a sharp transition at $\Theta = 0.60$ (TPR drops from 0.883 to 0.41). Proven by (E2); paper Figure 6.
- (C5): The IDM mitigation layer bounds ego braking to -4.0 m/s², reducing mean braking severity by approximately 75 % (mean bounded deceleration approximately -2.03 m/s²). Proven by (E1) via the `mitigated_a` column; paper Table 5.

A.4.2 Experiments

The full reviewer path takes approximately 1 h 55 min on a 16-core host; a minimum-validation path (install, smoke test,

E1, verifier) completes in approximately 30 min.

(E1): *Single- or five-seed reproduction.* `run_single_seed.sh 0` followed by `verify_results.py --seeds 0` runs the 18 detector-by-attack configurations and reports a PASS/FAIL verdict against the tolerance bands in `CLAIMS.md`, validating (C1), (C2), (C3), (C5); the five-seed protocol (`run_multi_seed.sh 0 1 2 3 4`, approximately 80 min more) reproduces the paper averages.

(E2): *Threshold sensitivity.* `analyze_multi_seed.py results/ sweeps  $\Theta$` over the E1 logs without recompilation and regenerates Figure 6, validating (C4).

Exact invocations, expected verifier output, per-claim tolerance bands, and the reference figures/tables for diff-based comparison are documented in `CLAIMS.md`, `reference_results/README.md`, and the repository docs/.

A.5 Notes on Reusability

The five installation stages run independently, and the detector factory is in `src/vasp/mdm/MisbehaviorDetectors.h` exposes a runtime-configurable `detectorType`; adding a detector needs only a `BaseDetector` subclass and a case in `createDetector()` (log schema, verifier, and figure pipeline unchanged), and adding a claim is a one-entry change in the `CLAIMS` table of `verify_results.py`.